MIL-F-51224B(EA)
18 September 1980
SUPERSEDING
MIL-F-51224A(MU)
21 March 1967

MILITARY SPECIFICATION

FILTERS, GAS-PARTICULATE, 600, 1200, 2500 AND 5000 CFM; ABC-M14A1, ABC-M15A1, ABC-M16A1 AND ABC-M17A1

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of gas-particulate filters for filter units.

1.2 Types. The gas-particulate filters shall be furnished in the following types as specified:

Type	Capacity (cubic feet per minute)
M14A1	600
M15A1	1200
M16A1	2500
M17A1	5000

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardiztion Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL5-19-1861 - Filter, Gas and Particulate, 600 CFM, ABC-M14A1.
TDPL5-19-1865 - Filter, Gas and Particulate, 1200 CFM, ABC-M15A1.
TDPL5-19-1850 - Filter, Gas and Particulate, 2500 CFM, ABC-M16A1.
TDPL5-19-1869 - Filter, Gas and Particulate, 5000 CFM, ABC-M17A1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on the applicable TDPL or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the filters shall conform to the specifications and drawings listed on the applicable technical data sheet.

3.2 Manufacture and assembly. The filters shall be manufactured, assembled and marked in accordance with the following drawings:

M14A1 - D5-19-1861
M15A1 - D5-19-1865
M16A1 - D5-19-1850
M17A1 - D5-19-1869

3.3 Preproduction. Prior to the start of regular production, a preproduction sample of gas-particulate filters of the same type shall be produced in accordance with this specification for examination and test (see 4.3).

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3.4 Workmanship. The filters shall be free from foreign matter (dirt, oil, and viscous material) and damage, such as distorted or cracked frames; deformation or sagging of medium separators, and faceguards; cracks in adhesive; and cracks and holes in exposed portions of the medium.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction.

4.3.1 Sample. A preproduction sample of:

four M14A1 gas-particulate filters,
two M15A1 gas-particulate filters,
two M16A1 gas-particulate filters, or
two M17A1 gas-particulate filters,

as specified in the contract shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2).

4.3.2 Inspection. Each gas-particulate filter in the preproduction sample shall be examined for all requirements of the applicable drawings and this specification.

4.3.3 Acceptance/rejection criteria. Each gas-particulate filter in the preproduction sample shall comply with the examination specified in 4.2.2 to be acceptable.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the gas-particulate filter of one type produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of particulate filters shall be represented in any one lot of finished gas-particulate filters.

4.4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample gas-particulate filters shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 Classification of defects.

(a) Filters, gas-particulate, 600, 1200, 2500, 5000 cfm (Drawing D5-19-1861, D5-19-1865, D5-19-1850 and D5-19-1869).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Component missing
	102	Component incorrectly assembled
	103	Marking incorrect
	104	Workmanship (3.4)
Minor:	AQL 2.5 percent defective	
	201	Marking illegible

(b) Packaging inspection (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 4.0 percent defective	
	101	Filter inlet and outlet openings not covered as specified
	102	Loose pack
	103	Liner missing or incorrectly applied
	104	Box closure incorrect
	105	Skids not as specified
	106	Strapping loose or missing
	107	Marking incorrect or illegible

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5. PACKAGING

5.1 Preservation - packaging.

5.1.1 Level A. Each filter unit, shall have each component filter, packaged separately in accordance with its Packaging Data Sheet, which is identified by its PS drawing number.

TABLE I. Filter components

Filter	Component
M14A1	Filter, Gas, 600 CFM Filter, Particulate 600 CFM
M15A1	Filter, Gas, 1200 CFM Filter, Particulate 1200 CFM
M16A1	Filter, Gas, 2500 CFM Filter, Particulate 2500 CFM
M17A1	Filter, Gas, 5000 CFM Filter, Particulate 5000 CFM

5.1.2 Commercial (particulate filters only). The particulate filters, (table I) shall be commercially packaged in accordance with their respective Packaging Data Sheets which are identified by their PS drawing numbers.

5.2 Packing.

5.2.1 Level A. Unit packages (5.1) shall serve as shipping containers.

5.2.2 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Level A. In addition to any special marking required by the contract, unit packages and shipping containers shall be marked in accordance with MIL-STD-129. Set marking for "Set or Assembly Marking" as specified in MIL-STD-129 shall be applied.

5.3.2 Commercial. Marking shall be in accordance with MIL-STD-1188, except that set marking for "Set or Assembly Marking" as specified in MIL-STD-129 shall be applied.

5.4 Repair parts. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair parts packaging data sheet which is identified by the National Stock Number for the repair part.

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6. NOTES

6.1 Intended use. The gas-particulate filters covered by this specification are components of filter units used for protecting fixed installations against toxic gases and particulate matter. The following is a list of gas-particulate filters and their respective filter unit assemblies.

Type	Assembly
M14A1	Filter Unit, Gas-Particulate, EMD, 600 CFM, ABC-M9A2.
M15A1	Filter Unit, Gas-Particulate, EMD, 1200 CFM, ABC-M10A2.
M16A1	Filter Unit, Gas-Particulate, EMD, 2500 CFM, ABC-M11A2.
M17A1	Filter Unit, Gas-Particulate, EMD, 5000 CFM, ABC-M12A2.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of gas-particulate filter required (see 1.2).
(c) Level of packing required.
(d) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contrat.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify contractor whether or not to proceed with production.

6.3 Packaging data sheet. The packaging data sheet is identified by the PS drawing number for each component filter and should be ordered from the contracting agency thereof.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 214. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A519

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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